

$$\frac{\partial r}{\partial m} = \underbrace{C = \frac{\partial A}{\partial m} u}_{\substack{\text{3rd order} \\ \text{tensor} \\ N_u \times N_u \times N_m}} \quad \text{Special yet frequent}$$

instead multiply Au , then take derivative to avoid computing 3rd order tensor

04/02/25

Lecture 20

min $\Phi(u(m), m)$, where u depends on m through solution $m \in \mathbb{R}^N$

of fwd model $r(u(m), m) = 0 \xrightarrow{A(u, m)} C(u, m)$

$$g(m)^T = \frac{\partial \Phi}{\partial m} + \frac{\partial \Phi}{\partial u} \frac{du}{dm} \Rightarrow \frac{du}{dm} = -A^{-1}C$$

$$\Phi(u, m) = \frac{1}{2} (Bu - d)^T (Bu - d) + \frac{\beta}{2} (m - m_0)^T R (m - m_0)$$

du/dm : direct sensitivity

$$g^T = \frac{\partial \Phi}{\partial m} + \frac{\partial \Phi}{\partial u} \frac{du}{dm}$$

$$\boxed{A^{-1}} \boxed{C}$$

Solve: $A \frac{du}{dm} = C$
 $N_u \times N_m$ $N_u \times N_m$

$$\beta R(m - m_0)$$

$$B^T (Bu - d)$$

$$P = -(A^{-1})^T \left(\frac{\partial \Phi}{\partial u} \right)^T$$

Solve: $A \left(\frac{du}{dm} \right)_i = C_i$
 each column

Adjoint method:

① solve $A^T P = - \left(\frac{\partial \Phi}{\partial u} \right)^T$ for $P \in \mathbb{R}^{N_u}$

adjoint state, co-state, dual variable

Solve $A^T P = - \left(\frac{\partial \Phi}{\partial u} \right)^T$

② $g = \frac{\partial \Phi}{\partial m} + C^T P$

→ given m , ① solve $r(u, m) = 0$ for u .

find $\frac{\partial \Phi}{\partial u}$, A . → then proceed to ②

Currently, we consider u as a function of m .
Instead w/ the Lagrangian framework (rather than adjoint), consider u, m as independent variables.

① construct Lagrangian

$$\mathcal{L}(u, m, p) := \Phi(u, m) + \underbrace{p^T r(u, m)}_{\text{scalar } \omega} \quad \begin{array}{l} \text{Lagrange multipliers } \in \mathbb{R}^{1 \times N_u} \end{array}$$

② Forward problem: $\frac{\partial \mathcal{L}}{\partial p} = r(u, m) = 0$

③ Adjoint problem: $\frac{\partial \mathcal{L}}{\partial u} = \left(\frac{\partial \Phi}{\partial u} \right)^T + A^T p = 0$

④ gradient: $\frac{\partial \mathcal{L}}{\partial m} = \left(\frac{\partial \Phi}{\partial m} \right)^T + C^T p = 0$

at a given m , solve for u w/ ②
then, with ③, knowing m, u , solve for p
finally, w/ ④, can solve for gradient

Lagrangian based adjoint method in inf-dim / continuous system

$$\mathcal{J}(m) = \min_{\substack{m \\ H^1}} \frac{1}{2} \int_{\Omega} (Bu - d)^2 dx + \frac{\beta}{2} \int_{\Omega} \nabla m \cdot \nabla m dx$$

$\downarrow$ observation operator $\uparrow$ from prior

 $\rightarrow$ step $\square$
or for points
 δ ; or Ω mini
gaussian w/
support

for ice sheet, we expect m
to be sharply changing
(streams $\rightarrow$ free slip, solid
ice $\rightarrow$ no slip) so, use
TV regularization instead

Consider an example problem to derive Lagrangian framework
 u depends on m through fwd model

$$-\nabla \cdot (m \nabla u) + \vec{v} \cdot \nabla u + au^3 = f \quad \text{in } \Omega$$

↗ nonlinear reaction term
 ↘ groundwater flow

(here)

u is chemical diffusion/
 subsurface flow

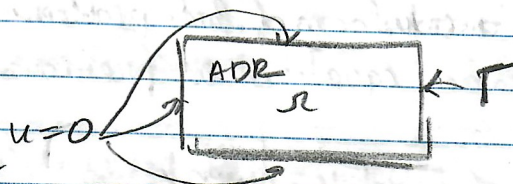
m is a spatially varying
 diffusion parameter

Note:

• can have 2 Φ or 2 unknown parameters
 → grad has 2 components

BC:

$$u=0 \text{ on } \Gamma$$



① Construct the Lagrangian
 (weak form of fwd problem)

cannot solve for an
 energy function,
 due to the advection
 term & cubic u .

$$\int_{\Omega} p(x) [-\nabla \cdot (m \nabla u) + \vec{v} \cdot \nabla u + au^3 - f] dx = 0 \quad \forall p \in H_0^1$$

↗ test function / adjoint variable

Integrate by parts to get weak form of fwd problem

$$\int_{\Omega} (m \cdot \nabla u \cdot \nabla p + p \vec{v} \cdot \nabla u + p a u^3 - p f) dx = 0 \quad \forall p \in H_0^1$$

$-\int_{\Gamma} p \nabla u \cdot \vec{n} ds$ to get weak form, try to get a $\nabla u \cdot \vec{n}$ term
 (Neumann / natural BC)

$$\mathcal{L}(u, m, p) = \frac{1}{2} \int_{\Omega} (Bu - d)^2 dx + \frac{\beta}{2} \int_{\Omega} \nabla m \cdot \nabla m dx$$

i.e:
 $\Phi(u, m)$

i.e:
 $\mathcal{P}^T r(u, m)$
 in weak
 form

$$\left\{ \begin{aligned} & + \int_{\Omega} (m \cdot \nabla u \cdot \nabla p + p \vec{v} \cdot \nabla u + p a u^3 - p f) dx \end{aligned} \right.$$

$$L(u, m, p) = \frac{1}{2} \int_{\Omega} (Bu - d)^2 dx + \frac{\beta}{2} \int_{\Omega} \nabla m \cdot \nabla m dx + \int_{\Omega} (m \nabla u \cdot \nabla p + p \nabla u \cdot \nabla u + p \alpha u^3 - p f) dx$$

② Variation of L w.r.t $p \Rightarrow$ weak form of fluid problem:

$$\delta_p L = 0 \quad \forall p \in H^1_0$$

$$\Rightarrow \int_{\Omega} (m \nabla u \cdot \nabla \hat{p} + \hat{p} \nabla u \cdot \nabla u + \hat{p} \alpha u^3 - \hat{p} f) dx = 0$$

$\hookrightarrow$ weak form of fluid problem

solve for $\hat{p}$ vs. $p \Rightarrow$ same equation as previously derived
 u , given m

③ Variation of L w.r.t $u \Rightarrow$ weak form of adjoint problem

$$\delta_u L = 0 \quad \forall \hat{u} \in H^1_0$$

$$\delta_u L = \int_{\Omega} \left[\underbrace{(Bu - d)}_{\frac{\partial \Phi}{\partial m}} \underbrace{B \hat{u}}_{\text{ATP}} + p \nabla \cdot \hat{u} + 3 \alpha u^2 \hat{u} \right] dx = 0 \quad \forall \hat{u} \in H^1_0$$

weak form of adjoint problem

Solve for $\hat{u}$ linear in p (as in strong form)

p , given u, m

④ Variation of L w.r.t $m \Rightarrow$ Fréchet derivative
 or, "weak form" of gradient

$$\delta_m L = 0 \quad \forall \hat{m} \in H^1$$

find gradient

based on

u, m, p

$$\Rightarrow \beta \int_{\Omega} \nabla m \cdot \nabla \hat{m} + \hat{m} \nabla u \cdot \nabla p dx = 0$$

$\equiv \langle G_1(m), \hat{m} \rangle$, directional derivative

Strong form expressions:

FWD problem: reverse integrate by parts [FILL IN]

$$\boxed{\begin{aligned} -\nabla \cdot (m \nabla u) + \vec{v} \cdot \nabla u + a u^3 &= f \quad \text{in } \Omega \\ u &= 0 \quad \text{on } \Gamma \end{aligned}} \quad \text{strong fwd}$$

Adjoint

$$\int_{\Omega} \left[(Bu-d) B^* \hat{u} + m \nabla \hat{u} \cdot \nabla p + p \vec{v} \cdot \nabla \hat{u} + 3 p a u^2 \hat{u} \right] dx = 0 \quad \forall \hat{u} \in H_0^1$$

integrate by parts to free $\nabla \hat{u}$

$$\int_{\Omega} \hat{u} \left[B^* (Bu-d) - \nabla \cdot (m \nabla p) - \nabla \cdot (p \vec{v}) + 3 p a u^2 \right] dx$$

$$+ \int_{\Gamma} m \nabla p \cdot \vec{n} + p \vec{v} \cdot \vec{n} \, ds = 0 \quad \forall \hat{u} \in H_0^1$$

($\hat{u} = 0$ on Γ)

$\hat{u}$ is arbitrary: (B^* : adjoint (or transpose) of B in inf-dim)

$$\Rightarrow B^* (Bu-d) - \nabla \cdot (m \nabla p) - \nabla \cdot (p \vec{v}) + 3 p a u^2 = 0$$

$$\boxed{\begin{aligned} -\nabla \cdot (m \nabla p) - \nabla \cdot (p \vec{v}) + 3 p a u^2 &= -B^* (Bu-d) \quad \text{in } \Omega \\ \text{Since } p \in H_0^1, \quad p &= 0 \quad \text{on } \Gamma \end{aligned}} \quad \begin{array}{l} \text{Strong} \\ \text{form of} \\ \text{adjoint} \end{array}$$

Properties:

• linear in p

• $A^T = -\nabla \cdot (m \nabla \cdot) - \nabla \cdot (\vec{v} \cdot) + 3 a u^2$ (operator)

Note: $-\nabla \cdot (p \vec{v}) = -\vec{v} \cdot \nabla p - p \vec{v} \cdot \nabla$; flow in reverse direction

leads to a non-self-adjoint A ; $3 a u^2$ is a linearization of $a u^3$

$\nabla \cdot (m \nabla p)$ leads to ^{self}-adjoint part of A

③ Gradient in strong form

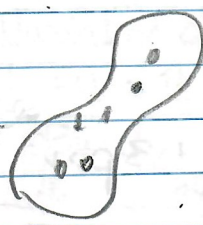
$$\beta \int_{\Omega} \nabla m \cdot \nabla \hat{m} \, dx + \int_{\Omega} \hat{m} \nabla u \cdot \nabla p \, dx = 0 \quad \forall m \in H^1$$

(weak form of gradient)

$$\int_{\Omega} (-\beta \Delta m + \nabla u \cdot \nabla p) \hat{m} \, dx + \int_{\Gamma} \beta \hat{m} \cdot \nabla m \cdot \vec{n} \, dS = 0$$

Since $\hat{m}$ is arbitrary on Ω & on Γ :

$$\Rightarrow \begin{matrix} \text{enforce} \\ -\beta \Delta m + \nabla u \cdot \nabla p \end{matrix} (=0) \left\{ \begin{matrix} \text{gradient on } \Omega \end{matrix} \right.$$



$$G(m) = \begin{cases} -\beta \Delta m + \nabla u \cdot \nabla p & \text{in } \Omega \\ \beta \nabla m \cdot \vec{n} = 0 & \text{on } \Gamma \end{cases}$$

↓

drive to 0

choose a BC to prescribe

↓

m is not changing in normal direction
or, have a dirichlet BC on m on Γ